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Ecological and socio-economic factors influencing aguaje (*Mauritia flexuosa*) resource management in two indigenous communities in the Peruvian Amazon

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ABSTRACT

In this study, we sought to better understand the social, economic, and ecological factors influencing the development of sustainable management practices for *Mauritia flexuosa* (aguaje) in two Maijuna indigenous communities. Focus groups, semi-structured interviews and household surveys were conducted to document current and historical patterns of aguaje harvest and management, the importance of aguaje to household income, and to identify current harvesting strategies and key perceptions regarding management. This was complemented with ecological sampling in 12 aguaje stands to determine population abundance, population structure, and sex ratio. Results indicated that *M. flexuosa* stands show clear evidence of past over-exploitation, with a current sex ratio of 3.48 males for each female, and low adult female densities (21 females/ha). Despite this, *M. flexuosa* remains highly abundant and populations maintain an inverted J-curve structure; however, results also showed a significant relationship between low densities of adult females and low densities of seedlings ($R^2 = 0.476$; $p = 0.013$), suggesting that the felling of females from the canopy is reducing recruitment rates. The current state of *M. flexuosa* populations corroborates the harvest history as told by interviewees that intensive destructive harvesting began in the 1990s and has since waned, primarily due to the decline in the stock of females and low prices. Attempts at non-destructive harvesting, via the use of climbing harnesses has had mixed success as nearly half of the *M. flexuosa* palms harvested in 2010 were cut down rather than climbed. Scarcity of adult females and low prices resulted in low return on effort, and current income from aguaje accounts for just 5% of the communities' cash income. Results indicated that four factors synergistically interact to hinder development of sustainable aguaje management: (1) low resource stock, (2) market barriers, including poor market access and low prices, (3) limited access to climbing technology and training, and (4) limited organizational experience at a community level, which limits the effectiveness of attempts to mitigate the first three barriers. Despite these challenges, the Maijuna and aguaje have many attributes associated with successful management of natural resources. Our study suggests that in cases where forest resources have already been degraded, transitioning to more sustainable management alternatives can be difficult as a number of ecological and socio-economic factors interact to hinder implementation.

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1. Introduction

Non-timber forest products (NTFPs) are often promoted as part of integrated conservation and development projects because they allow local communities to participate in market economies while keeping forests largely intact. Conservation organizations often presume NTFPs to be less damaging than other land use alternatives, such as logging, agriculture and rearing livestock, but recent studies have acknowledged they are not immune to overexploitation (Svenning and Macia, 2002; Ticktin, 2004; Endress et al., 2006; Martinez-Ramos et al., 2009; Lopez-Toledo et al., 2011). Harvesting of NTFPs often occurs with limited or no research conducted on the distribution and abundance of the resource, sustainable harvest rates, or the ecological effects of current harvest regimes. Also, NTFPs are often common-pool resources, which tend to be overexploited with a lack of management.

In the Peruvian Amazon, *Mauritia flexuosa* L.f., is a prime example of an NTFP experiencing localized overexploitation. Though this dioecious tropical palm, locally known as *aguaje*, has high ecological, cultural and economic value, destructive harvesting practices to obtain fruit undermine the palm's potential in rural and regional economies (Padoch, 1988; Delgado et al., 2007). Palm swamps dominated by *M. flexuosa*, known as *aguajales* in Peru, occur throughout the Amazon and make the resource accessible to

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harvesters in large quantities. Harvesters usually collect aguaje fruit by cutting down adult female palms, a practice thought to be ecologically and economically unsustainable. Demand for aguaje fruit in the urban center of Iquitos, Peru is strong, and the fruit is consumed raw or processed to make beverages, ice cream, and other products. Fruit is sold in 30–40 kg sacks containing 800–1200 individual fruit (Padoch, 1988; Delgado et al., 2007). Delgado et al. (2007), estimated 124 sacks/day are sold in Iquitos, which translates into a consumption rate of >1 million fruit/day. This demand pushes harvesters deeper into aguajales and into more distant watersheds, resulting in resource degradation before management plans can be developed. As fruit producing female palms are depleted, local people lose a source of income and are left with forests dominated by unproductive male palms (Vasquez and Gentry, 1989; Holm et al., 2008; Manzi and Coomes, 2009).

Overexploitation of *M. flexuosa* has been a cause of concern for decades, though little monitoring or research has been conducted. In 1989, Vasquez and Gentry described the conversion of expansive aguajales near Iquitos to stands of male palms, noted the increasing distances harvesters travel to obtain fruit and described declines in the availability of the fruit in Iquitos markets. To date, quantitative studies on the ecological effects of harvesting have been limited: Kahn (1988) reported the effects of harvesting on palm sex ratio and abundance in three plots across an expansive aguajal and beyond this, only a few studies have been conducted in harvested areas (Peters et al., 1989; Kahn and de Granville, 1992).

More recent studies have focused on *M. flexuosa* conservation, harvest and management techniques. In the Ecuadorian Amazon, Holm et al. (2008) used population matrix modeling to explore sustainable harvest rates of destructive harvesting methods and estimated that 15% of females could be cut every five years and that populations must maintain at least 20 females/ha in order to maintain positive population growth rates. Such a low rate of harvest may not be economically feasible and local communities often lack the time or ability to monitor harvest rates. Thus, most conservation efforts have focused on the development of non-destructive harvesting approaches. A handful of communities in the Peruvian Amazon have adopted such strategies, which may include the use of climbing devices to harvest fruit without felling and/or the cultivation of *M. flexuosa* in agroforestry systems or plantations (Bejarano and Piana, 2002; Delgado et al., 2007; Manzi and Coomes, 2009). Climbing devices allow harvesters to collect fruit without killing palms and thus maintain the production potential of those palms for future harvests. With the limited success of cultivation and slow spread of agroforestry systems containing aguaje palms (Delgado et al., 2007), harvesting *M. flexuosa* from wild stocks will most likely continue to be the primary source of commercial fruit.

However, the success of non-destructive management plans for wild populations has rarely been evaluated, with only the study by Manzi and Coomes (2009) addressing the issue. In this study, the authors concluded that the adoption of sustainable management activities (non-destructive harvest and agroforestry) depended on the prevailing conditions in the community, access to climbing devices, demonstrated effectiveness of these devices, and socio-economic characteristics of households. Moreover, they found that use of climbing devices stimulated the adoption of community-level palm conservation efforts as well as the incorporation of aguaje into agroforestry systems. While this study provided valuable insight, it remains unclear if their conclusions are applicable across the range of conditions faced by aguaje-harvesting communities in the Peruvian Amazon. Therefore, additional research is needed to more fully understand how a range of cultural, socio-economic, and ecological factors interact to influence the sustainable harvest and management of aguaje within communities. This information can provide valuable insight to better inform the design and implementation of community-based aguaje resource management activities.

In this study, we sought to better understand the social, economic, and ecological factors influencing the development of sustainable management practices of aguaje in the Yanayacu River basin of the Peruvian Amazon, an area inhabited by the Maijuna indigenous group. The Maijuna are interested in transitioning to sustainable aguaje management as part of a broader effort to develop sustainable economic options for their communities. On the surface, these communities appear to share some characteristics of the community described by Manzi and Coomes (2009); yet they are only seeing limited success in transitioning to non-destructive harvesting, suggesting that additional factors beyond those identified by Manzi and Coomes (2009) may influence the successful implementation of aguaje management.

To identify and better understand the critical factors affecting sustainable management of aguaje in this area, we (1) conducted semi-structured interviews and focus groups to document current and historical patterns of aguaje harvest and management, and participation in the regional aguaje market, (2) utilized household surveys and semi-structured interviews to determine the importance of aguaje to household income as well as to identify current harvesting strategies and key perceptions of aguaje resource management, and (3) sampled aguaje stands to evaluate the abundance, population structure, and sex ratio of aguaje stands. With this information, we identified key challenges facing further development of aguaje extraction and provide recommendations for resource management, considering the need for both ecological and socio-economic stability.

2. Aguaje and aguajales

A long-lived, arborescent and dioecious palm, *M. flexuosa* grows in aquatic and swamp habitats throughout the lowland tropics of South America. *M. flexuosa* is especially common in the Amazon Basin where it occurs predominately in high density swamp forests occurring in flood plains of rivers and streams or in poorly drained shallow depressions in upland forest that are flooded only by rains (Kahn, 1991). Young palms lack a visible aboveground stem, with persistent petiole bases masking the formation of a soft trunk. As the palm grows, a smooth, hard, and permanent trunk forms and is exposed as the petiole bases fall. Diameters reach 30–60 cm in adults, supporting stems that can exceed 30 m in height. The stiff costapalmate leaves, numbering between 8 and 20 on mature palms, measure up to 2.5 m long and 4.5 m wide. Inflorescences up to 2 m in length emerge from between petioles and support 25–40 branches of flowers. In females, the inflorescences become pendulous with fruit, often hanging with 4–5 dead leaves that may persist on the palm (Henderson et al., 1995). In the Peruvian Amazon near Iquitos, fruiting peaks between July and September (Vasquez and Gentry, 1989), though the fruiting season varies throughout its range.

Aguajales play several important ecological roles particularly in terms of providing habitat and food resources to wildlife as well as in carbon storage. In Peru, at least seven species of parrots use standing dead *M. flexuosa* palms for nesting (Brightsmith, 2005). The fruit also provides an important food source to these parrots, as well as to ungulates (e.g. lowland tapirs), primates (e.g. red howler monkeys), and fish, which in turn, are important seed dispersers and predators (Bodmer 1991; Henderson et al., 1995; Zona, 1999; Beck, 2006). Additionally, aguajales store a significant amount of carbon within their waterlogged soils and thick layers of organic matter largely composed of fallen palm fronds and inflorescences. Researchers working in the Orinoco River Basin found that aguajales store up to 95.91 kg/m^2 of carbon, underscoring the importance of these areas for carbon storage (Vegas-Vilarrubia et al., 2010).

Adding to the significance of aguajales, many communities, especially indigenous communities, obtain more than just aguaje fruit from these areas. Many of the animal species mentioned above, as well as various plant species found in aguajales, are used for cultural, subsistence and economic activities (Gilmore, 2005). Additional recognized uses of aguaje include not only fruit consumption, but also oil, fiber, medicines, and construction material (Mejia, 1988). Stems of felled *M. flexuosa* are also used to cultivate palm beetle larvae (*Rhynchophorus palmarum*), locally known as *surí*, which is both a subsistence and market product (Padoch, 1988).

3. Study area

Fieldwork was conducted during August 2010 and May 2011 in the Maijuna communities of Puerto Huamán and Nueva Vida, located along the Yanayacu River in the northeastern Peruvian Amazon (Fig. 1). The Maijuna are also known as the Orejón (Steward, 1946; Bellier, 1993, 1994). Currently, there are approximately 400 Maijuna individuals living in four communities along the Yanayacu, Algodón, and Sucusari Rivers. These river basins, comprised of over 300,000 ha of intact primary rain forest, are part of their ancestral territory (Gilmore et al., 2010). This is a relatively flat area, similar to the rest of the Peruvian Amazon, with elevations ranging from 80 to 200 m a.s.l. (Vriesendorp and Foster, 2010). The area includes both upland and floodplain forests. Aguajales range in type and size, yet current harvest largely occurs in aguajales located in upland forest depressions flooded by rains. The region is tropical, warm, and humid, with a mean annual temperature of 26 °C and a mean annual precipitation of approximately 3100 mm per year (Marengo, 1998).

The communities of Puerto Huamán and Nueva Vida have populations of 95 and 109 people, respectively, the majority of which are of Maijuna descent. The two communities are 4 km apart along the Yanayacu River, which is 220 km by river from Iquitos, the largest city and commercial center in the region (Fig. 1). However, the trip is shortened to 95 km by crossing the thin isthmus between the Napo and Amazon Rivers by road at Mazán, a small port town. The Maijuna rarely travel to Iquitos but do participate to varying

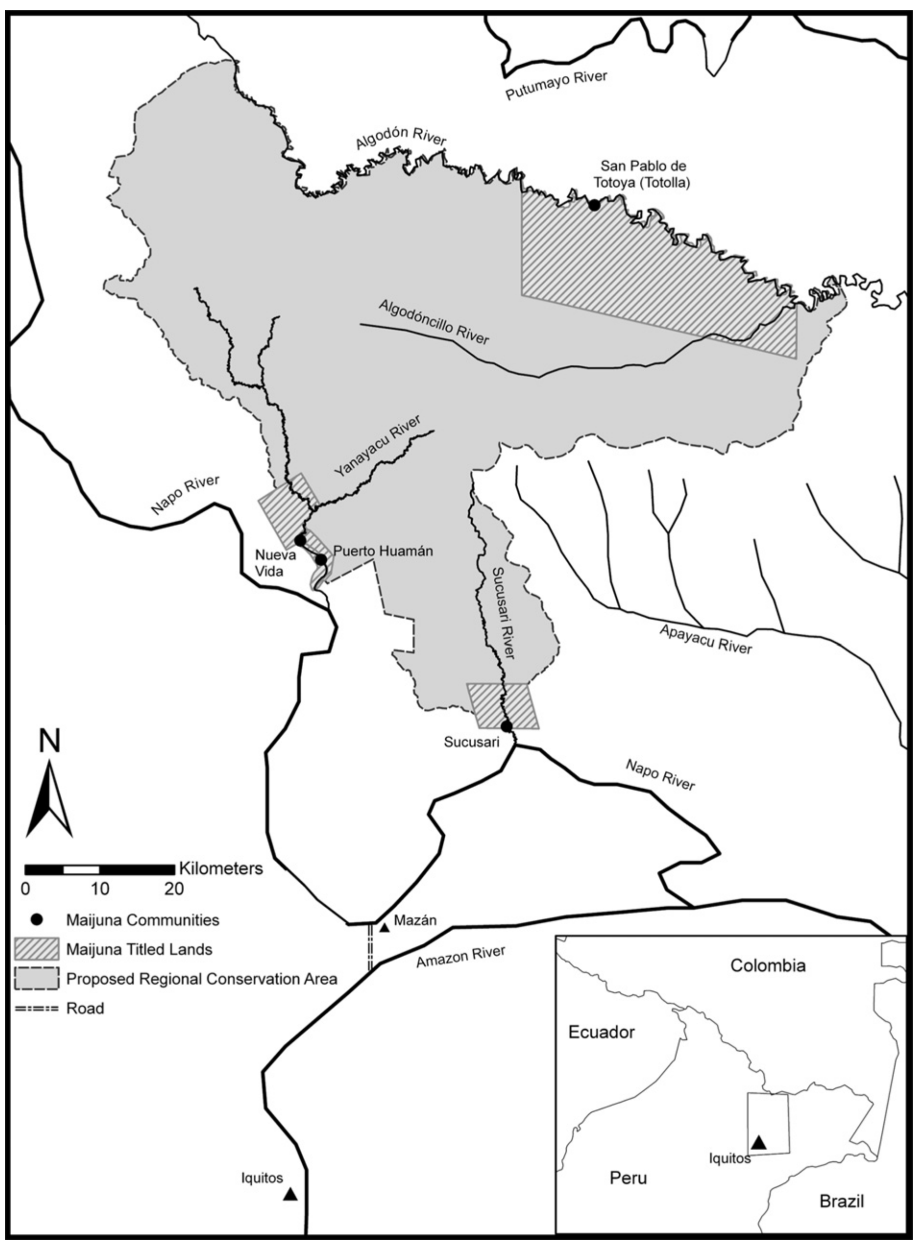


Fig. 1. Map of the study area. Research was conducted along the Yanayacu River, within and near the Maijuna villages of Puerto Huamán and Nueva Vida.

degrees in the market economy by conducting business with neighboring communities, in Mazán, or with middlemen working between villages and the Iquitos and Mazán markets. Economically poor, the communities rely heavily on natural resources and employ a variety of subsistence and income-generating strategies, including hunting, fishing, swidden-fallow agriculture, and the gathering of various forest products, including aguaje fruit (Gilmore, 2010).

The Maijuna have recently (2008) halted logging on their titled and traditional lands and have increased their interest in other, more sustainable, economic activities including NTFP extraction. Additionally, the Maijuna communities are pushing for the creation of an *Área de Conservación Regional* (ACR; regional conservation area) encompassing over 336,000 ha of their ancestral land to which they lack title (Gilmore et al., 2010). Protection of this land and a desire to show government officials they are effective land stewards has increased their desire to find sustainable economic activities and develop and implement forest resource management plans. Community leaders have expressed particular interest in *M. flexuosa* management and increasing their participation in the established regional market.

4. Methods

4.1. History of aguaje harvest and management

One of the authors (Gilmore) has worked with the Maijuna for over a decade on a variety of community-based biocultural conservation projects, has established social ties within each of the communities and has extensive knowledge of Maijuna resource use and income generating strategies. Harnessing these relationships and experiences, we conducted focus groups and semi-structured interviews with community experts and leaders regarding the history of aguaje fruit extraction and markets, aguaje resource management, and changes in aguaje harvest and management through time. Additionally, employees of a local conservation non-governmental organization (NGO) who have worked with the Maijuna since 2008 on natural resource and conservation issues were interviewed regarding more recent aguaje management and marketing efforts. To identify key themes, issues, and perceptions, the data were coded, organized and analyzed based on the methods described by Strauss and Corbin (1998).

4.2. Current harvest patterns and contribution to household income

To document current aguaje harvest and management, we conducted structured surveys with the heads of households, or their spouses, in the two communities. All 20 households in Nueva Vida and 16 out of 17 households in Puerto Huamán were surveyed. Surveys provided data on household demographics, mode of aguaje harvest, frequency of harvest, the economics of aguaje harvest and the underlying factors influencing patterns and intensity of aguaje harvest in 2010. Qualitative data collected were coded, organized and analyzed (Strauss and Corbin, 1998). To further investigate these topics and clarify necessary items we conducted follow-up semi-structured interviews in May 2011. At this time we also conducted structured surveys to measure household income earned from May 2010 to April 2011. Notably, this is from the start of one aguaje harvesting season to the next, a time period easily defined and understood by survey participants. Documentation of income generating strategies was based on extensive knowledge of Maijuna culture and livelihood strategies (Gilmore, 2005, 2010), and the survey was designed to capture the full range of income generating strategies employed throughout the communities. Maijuna leaders reviewed the survey for breadth and accuracy. Surveys were conducted based on participant recall with all numbers cross-checked with interviewees. Surveys were designed to document household cash income; quantifying resources generated from subsistence activities was beyond the scope of this project.

4.3. Aguaje abundance, population patterns, and sex ratios

We assessed the current state of *M. flexuosa* as a resource for the Maijuna by collecting individual palm and population-level data in 12 aguaje stands near the two communities. Aguajales were identified and mapped in a previously conducted participatory mapping project (Gilmore and Young, 2010). Using Maijuna maps and guides, we selected stands of varying accessibility to the communities in aguajales occurring in poorly drained depressions in upland forest that are inundated only by rains. All sites were within one half-day of travel (by foot and/or boat). To capture variability within the palm swamps, plots within each stand consisted of three circular subplots, each with 10.3 m radii, resulting in an area sampled of 0.1 ha per stand. Subplots were spaced at least five meters apart. To determine the location of the plots within the aguajales, we first determined the perimeter and extent of the aguajal and then established the plots in areas representative of the aguajal in terms of forest structure and composition. No plot was within 10 m of the edge of the aguajal.

For each plot, we collected data on *M. flexuosa* abundance, size class distribution, recruitment, evidence of recent harvest and adult sex ratios. Within each plot, all *M. flexuosa* stems $\geqslant$0.5 m height were counted and measured. Heights were measured to the highest point of leaves, directly for all individuals with a height <2 m and estimated using a clinometer for all others. We determined the sex of adult individuals by identifying inflorescences either attached at the tree crowns or fallen at the base of the palms. The sex of palms with no inflorescence and/or adjacent to another adult palm (thus making it unclear as to the source of fallen inflorescences) was recorded as unknown. Due to their much higher densities, seedlings <0.5 m height were identified and counted in four nested 2 m^2 plots located systematically across each circular subplot.

Using the data collected, we classified individuals into the following size-classes based on height: 0–0.99, 1.0–2.99, 3.0–5.99, 6.0–9.99, 10.0–19.99, 20.0–29.99, and $\geqslant$30.0 m. To aid in comparisons between other *M. flexuosa* studies, size classes were similar to Holm et al. (2008) and density data were calculated and reported on a per hectare basis. Assuming an approximate 1:1 sex ratio for unharvested populations, as was reported by Kahn and de Granville (1992), we used a Fisher's exact test to test for differences between the pooled sex-ratio of the plots and the 1:1 sex ratio of a healthy population of the same density.

5. Results

5.1. History of aguaje harvest and management

Within the Yanayacu River basin, aguaje fruit has only been harvested for the market economy for the past two decades. Previously, people collected fruit opportunistically from the ground and infrequently cut palms for subsistence fruit harvest and/or surí cultivation. During the early 1990s, large-scale commercial extraction began as people from outside the communities arrived to harvest and buy fruit. Commercial extraction included both outsiders and community members. Informants indicated that during this time aguaje was abundant. As one person stated, "In the past, in one day a boat could be filled with 40 or 60 sacks, filled in one day and only in one community." During this time up to 100 sacks per day were harvested. There was consensus among interviewees that a high volume of aguaje was annually harvested during this period, and it was easier to harvest due to the great abundance of easily accessible fruiting females – those occurring in aguajales near communities and rivers or streams.

The situation changed in the 2000s as commercial extraction declined as a result of overharvesting, with considerably fewer buyers and harvesters travelling to the area to harvest. As one individual explained, "When I was a child there was a lot more aguaje... there weren't people buying aguaje, there just weren't any. In approximately the year 1990, buyers from the outside came to look for and buy aguaje. And, people started cutting lots of aguaje and this is how things were destroyed." All harvesting of aguaje and other forest resources by outsiders came to a halt in 2008 when, in an effort to better conserve and manage their biological resources, the Maijuna actively restricted upriver access by non-community members.

In 2009, *Proyecto Apoyo al PROCREL* (PAP), a regional consortium working on biodiversity conservation and protected area management conducted several aguaje management workshops in the communities. They taught a limited number of harvesters to use climbing harnesses and helped the communities form aguaje management committees (*Comites de Aguaje*). Leaving a total of six harnesses for the two communities, it was hoped that (1) trained harvesters would teach others in their communities to climb, (2) the Comites de Aguaje would enact sustainable management practices for aguaje resources, and (3) the communities would be able to share the limited number of harnesses among community members. To further promote sustainable harvesting, PAP arranged for aguaje buyers in Iquitos to purchase aguaje from indigenous communities, including Nueva Vida and Puerto Huamán, at a value-added price with the condition that they be harvested non-destructively. Organization of these sales would fall under the purview of the Comites de Aguaje.

5.2. Current aguaje harvest and management

Results from our household surveys and interviews revealed that during the 2010 harvesting season, 72% of households collected aguaje, gathering a total of 204 sacks between the two communities (Table 1). This is considerably less than estimates given by community members from when commercial aguaje extraction was at its peak in the 1990s (Section 5.1). Based on our sampling of 15 sacks from Iquitos markets (mean number of fruit/sack = 865, SE = 27.3), this translates to approximately 176,473 fruit harvested. Harvest was usually conducted by husband and wife pairs and followed seasonal patterns of fruit production, with the majority of harvesting occurring between June and August.

Table 1
Summary data from household surveys.

	Households without climbers	Households with trained climbers
Number of households	19	17
Number of harvesting households (2010)	13	13
Average head-of-household age	42.5	36.6
Average household size	5.5	6.1
Total sacks collected	64	140
Average number of sacks collected per harvesting household	4.6	11.7
Number of palms cut down	27	20
Number of palms climbed	3[a]	45
% Palms cut of total palms harvested	90	31

[a] Households without climbers occasionally harvested with climbers from other households.

One year after the climbing training in 2009, destructive harvesting remained prevalent, as 49% of the aguaje trees harvested were cut down. However, the majority of trees (51%) were climbed (Table 1). Even households with trained climbers occasionally felled palms; while households with climbers used harnesses 69% of the time, they still cut 20 palms, which accounted for 43% of all cut palms in 2010 (Table 1). In total, nine households with climbers occasionally cut palms. On the other hand, households with no trained climbers destructively harvested 90% of the time, though in a few instances they received assistance from others who were trained (10%).

Surveys indicated that households received anywhere from US$1.09 to US$12.72 per sack of aguaje harvested, with an average of $3.86/sack. Price varied depending on when and to whom it was sold. Both buyers from the community and outside buyers offered similar prices (between US$1.09 and US$5.45). Visits from outside buyers were not as common as reported in the past, and only 25% of sacks were sold to external buyers.

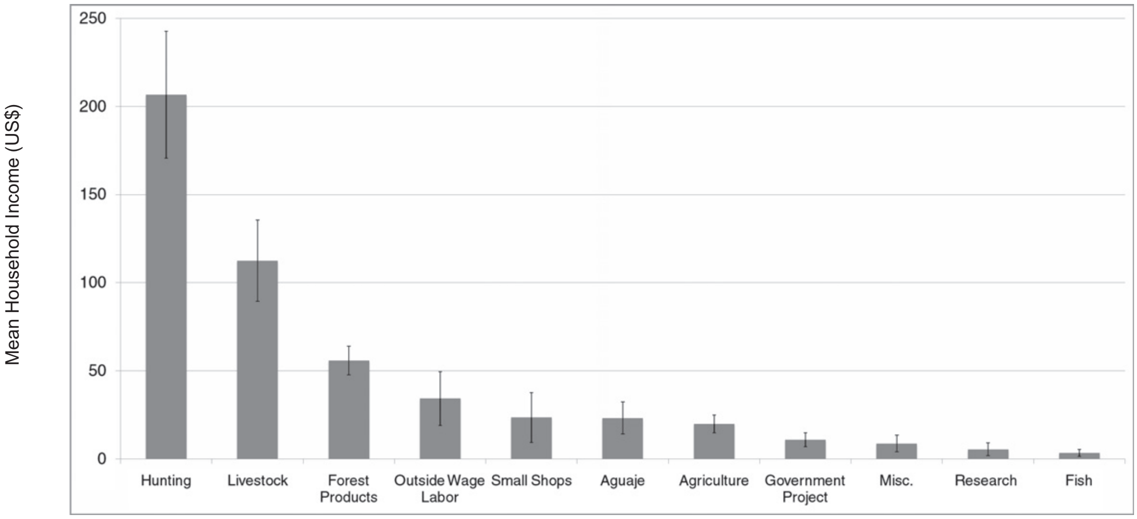
5.3. Factors affecting harvest intensity

Households provided a number of reasons for the relatively limited harvest of aguaje in 2010 as compared to harvest levels in the 1990s. All households interviewed (100%) indicated that there had been a decline in abundance, and many also noted the increased distance required to harvest. For example one harvester stated, "[In the past] Juan's (a pseudonym) brother took a 15 meter boat out of here filled with more than 90 sacks." This same individual also explained, "I harvested 8 sacks in one day. The work of one person was 8 sacks when there was a lot... At first it was close and now you have to walk an hour or an hour and 20 minutes into the center of the forest... There is aguaje but you have to carry it far..." Currently, households average 2.5 sacks per collecting trip. When asked the cause of the decline in aguaje, all respondents (100%) mentioned the cutting of females.

Despite the acknowledged decline in the stock of aguaje fruit, only 20% of interviewees directly cited the decline or increased effort required to harvest as reasons for not harvesting, or not harvesting more frequently during the 2010 season. However, 52% of respondents cited the low price offered for aguaje as a reason for not harvesting or harvesting less, which may be related to a decline in stock as the current prices are not seen as worth the effort now required to harvest. Higher prices offered by buyers seem to prompt people to travel the extra distance to harvest. One interviewee explained, "Sometimes when the price was 20 or 25 soles (US$7.27–US$9.09) per sack we would cut down 2 or 3 trees just to fill up one sack." Thus, harvesters evaluate the price of aguaje in relation to how much time and effort it takes to harvest and act accordingly.

Fifty-two percent of people also pointed to their inability to climb (due to a lack of knowledge or a lack of harnesses) paired with the desire to respect the communities' efforts to sustainably manage aguaje as reasons for not harvesting or not harvesting more in 2010. Other responses included being too busy with other work (12%), the lack of aguaje buyers coming upriver (12%), lack of transportation (4%), and lack of premium price contracts (4%). The last three responses (totaling 20%) reflect the difficulties in accessing the market and selling at prices harvesters felt reflected their investment in time and effort.

The current market chain has harvesters selling at low prices to either a buyer in the community or to a buyer from Mazán who periodically travels up the Yanayacu. These buyers then sell the purchased aguaje in Mazán. By the time they have reached the markets of Iquitos the sacks of fruit have passed through various



	Hunting	Livestock	Forest Products	Outside Wage Labor	Small Shops	Aguaje	Agriculture	Government Project	Misc.	Research	Fish
Income Contribution (% Community Total)	41	22	11	7	5	5	4	2	2	1	<1
Participation (% Households)	93	93	100	29	14	72	54	32	14	11	25

Fig. 2. Mean household income by source in the Maijuna communities of Puerto Huamán and Nueva Vida, May 2010 to April 2011. Error bars represent standard error. Outside Wage Labor refers to wages earned while working in Mazán, Iquitos, or otherwise outside the Yanayacu Basin. Forest Products refers to non-timber forest products other than aguaje and animal products (e.g. harvest of *Euterpe precatoria*). Research refers to wages earned working with research linguists and Government Project refers to wages earned while constructing a new community schoolhouse funded by the government.

hands, with the price per sack increasing with each exchange. The Comites de Aguaje have been tasked with organizing collective action to overcome some of these barriers, but interviewees felt the Comites need to be strengthened and better organized.

Though the Comites de Aguaje, with the help of PAP, organized harvesting groups in 2009 to fill orders at US$5.45 per sack for a particular Iquitos-based buyer (the premium price PAP helped negotiate), no aguaje was sold in this manner in 2010. However, this same price was achieved through middlemen on a couple of occasions. According to informants, the logistics of getting aguaje to Iquitos largely prevented Maijuna harvesters from taking advantage of the value-added price procured by PAP in 2010. For example, in 2009, Maijuna harvesters wound up spending any additional earnings on transportation resulting in little-to-no increase in income.

5.4. Factors affecting implementation of sustainable harvest techniques

The primary reason for the continued prevalence of destructive harvesting was that 19 out of 36 households (53%) did not have a member who participated in PAP's training or otherwise learned to use climbing harnesses. Additionally, interviewees indicated that training sessions were crowded and a limited number of trainers and harnesses restricted the amount of practical experience participants gained during the sessions. All households currently without trained climbers (100%) expressed interest in learning to climb, either by the heads-of-household or other household members. During interviews the need for more climbing training as well as more harnesses, preferably one for each household, were consistently mentioned.

Two-thirds of households who had trained climbers yet destructively harvested some females in 2010 stated they cut the palms because they were very tall and deemed too dangerous to climb. Other safety concerns included old trees, identified as ones that had drying crowns, as well as the presence of stinging ants (by one and two interviewees, respectively).

5.5. Contribution of aguaje to household income

Households in the communities relied on a wide range of activities to generate income, especially activities related to natural resource extraction. Household incomes, excluding those of the teachers, averaged $505 (range: $243–$1,000), and households participated in an average of five income-earning activities. Despite widespread participation in aguaje harvest (72%), it was not one of the principle sources of cash income for households (Fig. 2). The combined income from aguaje fruit totaled $650, which accounted for only 5% of the communities' total income (excluding teacher households). Only three households earned more than 10% of their total income from aguaje (range 0–26%). However, aguaje likely contributes indirectly to income generated by hunting as well (41% of community income), as aguajales are important hunting areas since many hunted species (paca, peccary, tapir, etc.) visit aguajales to consume aguaje fruit (Bodmer, 1991; Henderson et al., 1995; Zona, 1999; Beck, 2006).

5.6. Aguaje abundance, population patterns, and sex ratio

Densities of aguaje in the twelve stands sampled are reported in Table 2 and Fig. 3, with an average density of 5798.6 stems/ha (SE ± 933.5). Stands were dominated by seedlings (<1.0 m), which accounted for 71% of individuals recorded. Densities of non-seedling aguaje averaged 1662.5/ha (SE ± 182.5) and ranged from 630–2730/ha. There was an average of 93.3 (SE ± 8.9) reproductive adults/ha including 20.8 female stems/ha (SE ± 6.5).

The size class structure of each stand (Fig. 3) as well as the pooled size class structure (Fig. 4) revealed a population structure with an inverted J-curve, that is a high proportion of individuals in the two smallest size classes and a small proportion of palms in the larger size classes. As individuals increased in height, so did the proportion of reproductively active individuals in each size class moving from 0% (individuals <10m), followed by 27% (10–19.99 m height), 91% (20–29.99 m), and 100% for palms $\geqslant$30 m. However, five of the 12 stands had no individuals in the largest size class ($\geqslant$30 m).

Table 2
Observed size class and sex distribution across stands, given as number of individuals per hectare.

Category		Site 1	Site 2	Site 3	Site 4	Site 5	Site 6	Site 7	Site 8	Site 9	Site 10	Site 11	Site 12	Mean
Height range	Size Class													
< 1.0 m	1	1770	9810	6927	1123	6410	5330	557	1906	4850	4080	4813	2057	4137
1.0–2.99 m	2	860	1440	2060	540	1010	1090	900	900	1620	1120	1630	450	1135
3.0–5.99 m	3	360	430	420	170	330	450	370	190	250	310	560	40	323
6.0–9.99 m	4	80	90	110	30	40	70	90	10	10	170	110	10	68
10.0–19.99 m	5	20	120	50	20	–	10	40	50	10	130	120	20	49
20.0–29.99 m	6	80	100	70	70	50	70	70	80	90	30	100	100	76
⩾30 m	7	10	–	20	–	–	30	–	20	10	–	30	10	11
Total		3180	11990	9657	1953	7840	7050	2027	3157	6840	5840	7363	2687	5799
Cut stumps		40	30	–	–	–	10	10	40	40	–	10	20	17
Male		80	90	50	70	40	80	70	90	60	60	80	100	73
Female		–	70	20	–	10	10	10	10	30	20	60	10	21

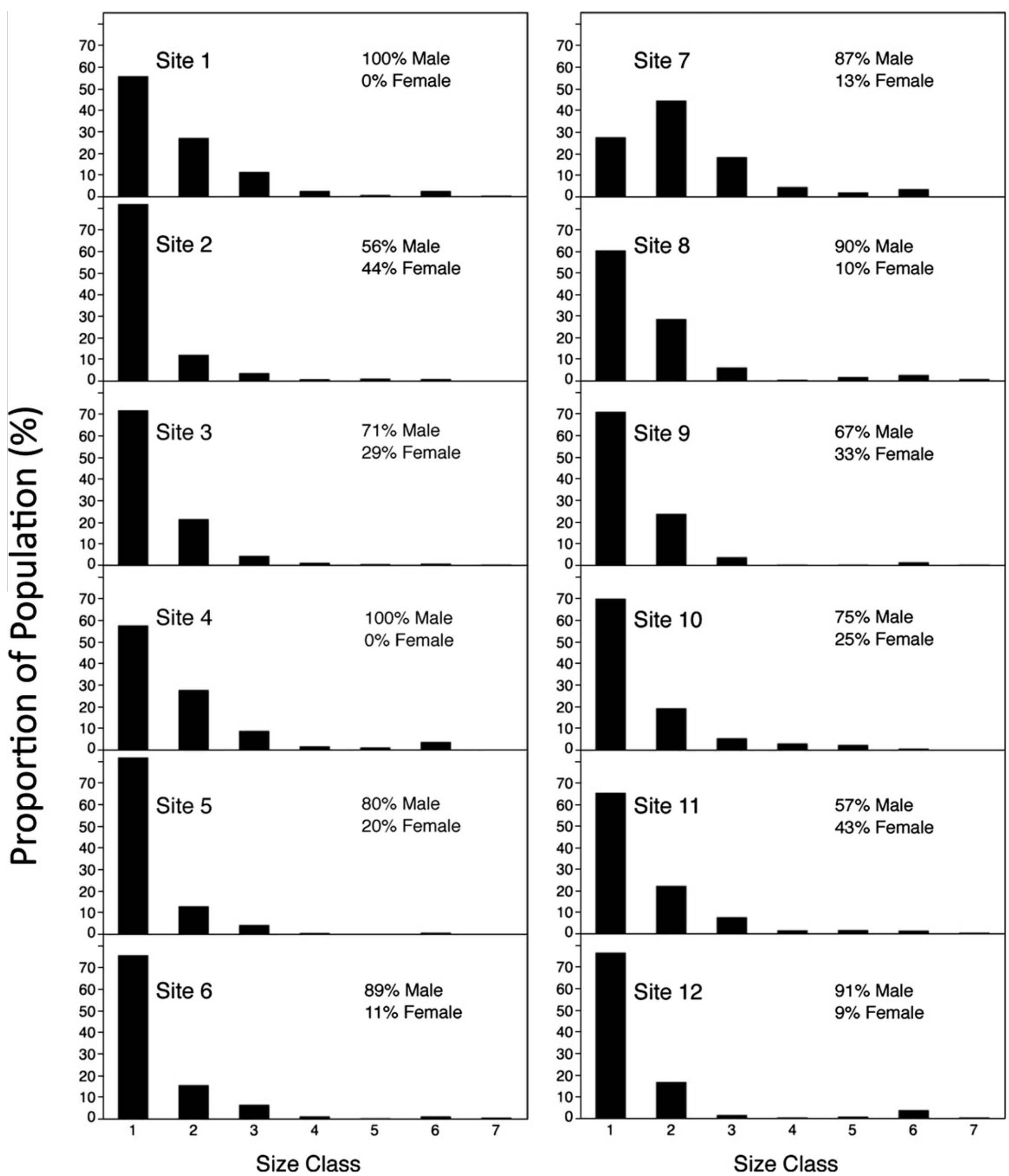


Fig. 3. Proportion of *M. flexuosa* size classes and stem density per stand and percent of known male and female reproductive adults. See Table 2 for absolute densities.

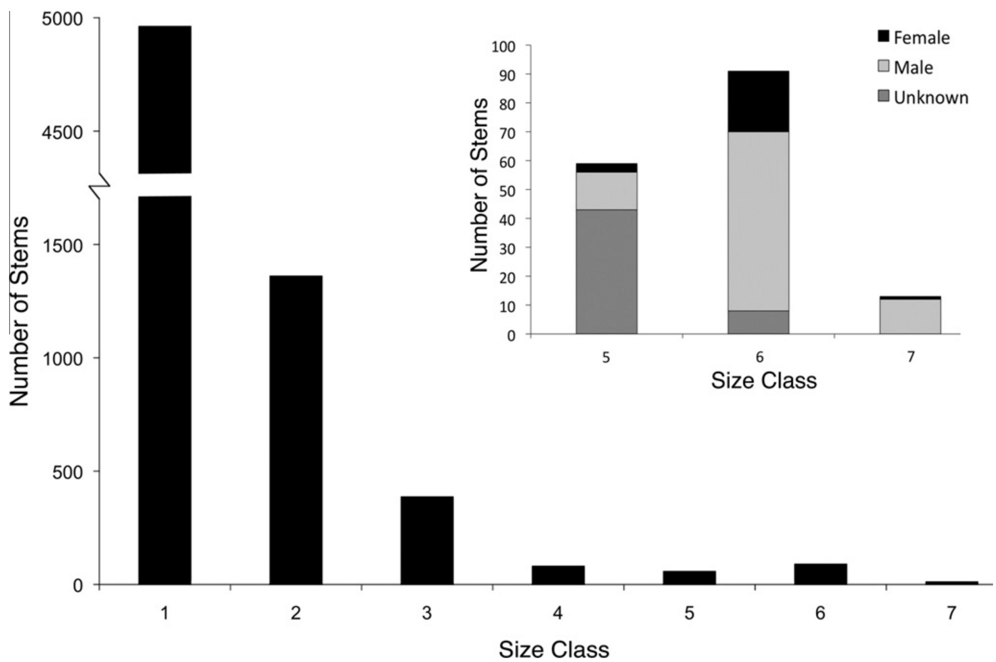


Fig. 4. Pooled size class structure of *M. flexuosa* sampled in the Yanayacu River basin. Total stems pooled across the entire area sampled (1.2 ha).

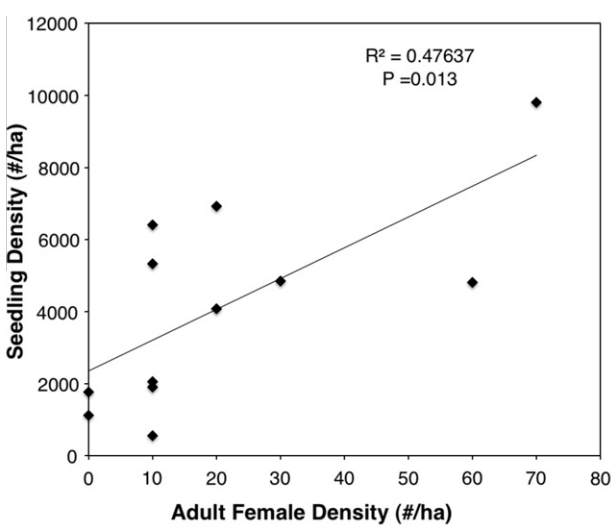


Fig. 5. Linear regression showing the relationship between the density of adult female *M. flexuosa* individuals and seedling (<1.0 m height) density in the twelve sampled plots.

Reproductive males outnumbered reproductive females by a considerable margin (Table 2 and Fig. 4 inset). For our study area, we report an overall sex ratio of 3.48:1 male to female ratio, with a density of 20.8 females/ha. This ratio was significantly different from 1:1 ($p < 0.0001$). Sex ratios varied among the stands ranging from close to 1:1 (plots 2 and 11; Table 2) to 10:1 (plot 12). Two stands had no recorded females, though females were observed outside of our plots within these aguajales at very low densities. In each of the stands sampled, we encountered stumps from recently cut females, though not all sampled plots contained such stumps (8 out of 12; Table 2).

Results from a linear regression also showed a significant relationship between the female density in the overstory and the density of seedlings (Fig. 5; linear $R^2 = 0.476$; $p = 0.0130$). As adult female density declined, so did seedling densities suggesting that felling of females may be affecting aguaje recruitment rates.

6. Discussion and conclusions

Through focus group discussions, interviews, household surveys and ecological sampling we were able to construct a comprehensive assessment of the status of aguaje as a socio-economic resource for two Amazonian indigenous communities. This multidisciplinary evaluation revealed several characteristics of the communities and their aguajales – social, economic and ecological – that hinder efforts to sustainably manage this resource.

6.1. Current status of aguaje resources

In the upland aguajales of the Yanayacu River basin, *M. flexuosa* populations seem to be approaching a critical juncture. While the total and adult stem densities for these populations fall within the range reported by Kahn (1992), adult female densities are 40% lower than those reported by Holm et al. (2008) for a low-intensity harvested population and comparable to the 23 females/ha found by Kahn (1988) in intensively harvested areas. Moreover, our reported adult female density is close to the 20 per ha density suggested by Holm et al. (2008) to be the minimum female density required to maintain positive population growth rates. The 3.48 sex ratio of males to females also points to the scarcity of females, and the trend in declining seedling densities associated with low densities of adult females suggests that in some areas recruitment rates have already been affected. In turn, reduced fruit production may negatively affect ecologically, economically and culturally important wildlife (Moegenburg and Levey, 2002), which can further impact the communities' ability to meet livelihood needs as hunting game is their primary source of cash income in addition to being important for subsistence. The inverted J-curves of the population size structures reported in our study are typical of long-lived species and *M. flexuosa* in particular (Holm et al., 2008; Bonesso Sampaio et al., 2008). In species with such population structures, the high density of seedlings ensures replacement of adult stems despite low survivorship. Although the populations were dominated by seedlings, inferring population trends from population size class structures is generally inappropriate, and more research is needed to better understand the impact of harvest and biased sex ratios on demography and long-term population dynamics. Despite the ecological and economic importance of *M. flexuosa*, there is surprisingly little known about its population ecology including seedling recruitment rates, establishment, growth, survival, and reproduction. The low density of female palms, missing females in several plots, and the unbalanced sex ratio indicate populations in the area have been severely degraded.

Regardless of the current risk to population growth rates, as the fruit-bearing females provide the marketable good, the decline in females effectively limits the economic value of aguaje as a natural resource. The low level of harvest reported during the 2010 season corroborates this conclusion. The 204 sacks collected by the community are a sharp contrast to the collection of up to 100 sacks during single harvesting events as reported in the past. At five percent of the communities' income, aguaje is currently not as major of an economic resource to the Maijuna as it has been in the past, or as they hope for it to be now and into the future.

6.2. Factors influencing sustainable management

With low harvest rates and a mixture of climbing and destructive harvesting techniques, the current status of aguaje harvest indicates that despite interest from the Maijuna and some limited NGO support, efforts to sustainably manage aguaje have encountered several challenges. Results indicate that four factors synergistically interact to hinder development of sustainable palm management: (1) low resource stock available to harvest, (2) market barriers, including poor market access and low prices, (3) limited access to climbing technology and training, and (4) limited organizational experience at a community level, which limits the effectiveness of attempts to mitigate the first three barriers.

Both the patchy nature of the upland aguajales occurring in the area and the decline in females due to destructive harvesting contribute to the low resource stock. As the number of females declines harvest effort increases, thus reducing the amount of income earned per unit of effort. This is compounded by the poor prices currently offered by buyers or the expense (and increased effort) of moving goods directly to Mazán or Iquitos. Furthermore, the current low economic return on effort hinders the implementation of sustainable management practices, because such practices add to the amount of effort required to harvest aguaje in the present (e.g. amount of time to climb, sharing harnesses, etc.), though they may also eventually reduce effort and increase income in the future.

With over half the households without trained climbers, and the limited amount of training and number of harnesses, the previous attempts to offer an alternative to destructive harvesting were insufficient. The Comites de Aguaje did not have enough organizational experience or authority to reinforce initial efforts to introduce climbing technology to households. Focused capacity building to strengthen community organization would have aided efforts to ban destructive harvesting, organize community transport of harvested aguaje and organize harness sharing and maintenance. Increasing access to climbing harnesses and training would ease the stress on the resource base, but not necessarily lead to aguaje serving as a sustainable income stream if the other factors are not addressed.

In their study of community-based aguaje resource management in the Peruvian Amazon, Manzi and Coomes (2009) pointed to the importance of several factors within the community that they worked in that contributed to the successful management of aguaje. Our study identified two additional factors: resource stock levels and market access. The community reported on by Manzi and Coomes (2009), Roca Fuerte, had recently been resettled from an area surrounded by degraded aguajales to a new location with more abundant aguaje, thus giving the community both the experience of scarcity and a new opportunity for sustainable management. But more than giving the community a fresh start in managing aguaje, the abundant stock provided a solid return on effort, despite the initial increased effort of management activities. By contrast, the Maijuna are trying to initiate changes in management in areas already degraded. Additionally, they are becoming used to aguaje scarcity and efforts to sustainably manage the low level of stock remaining find only moderate traction despite the considerable support for the idea. Limited market access also limits return on effort for the Maijuna. As aguaje extraction has declined in the Yanayacu, buyers from Mazán and Iquitos have been less regular and the need to go through several middlemen increases effort and lowers the amount earned. Though not discussed by Manzi and Coomes (2009), Roca Fuerte also benefits from reliable market access, as it is located along the Marañon River with consistent riverboat traffic transporting goods directly to Iquitos. Thus for that community, a robust resource stock and market access were ideal conditions to implement changes in aguaje management, while for the Maijuna these two factors are not only problematic but appear to be the main impediments to sustainable management.

Finally, the Roca Fuerte community appeared to benefit from years of sustained capacity building and support from an NGO, another difference from the communities studied here. NGO collaboration with the Maijuna communities is relatively new (with activities starting in 2008 and aguaje targeted work in 2009) and activities, including the training on aguaje resource management, have been limited in scope due to lack of funding.

6.3. Solutions to overcome challenges

The development of aguaje as a sustainable economic resource for the Maijuna communities will require more than simply adopting the use of climbing harnesses. Specifically the problems associated with low resource stock and market barriers must be addressed. Halting destructive harvesting is an important first step in addressing low stocks. However, since *M. flexuosa* takes years to become reproductive, additional measures to increase the resource base more quickly should be considered. Such efforts may take several forms. Incorporating aguaje into current agroforestry systems in addition to enhancement plantings within nearby degraded aguajales will increase the abundance of fruiting females while also decreasing the effort associated with harvesting. Additionally, as many aguajales in the area maintain populations with large proportions of seedlings and juveniles, the silvicultural technique of selective thinning may offer a relatively quick return on effort. In the case of aguaje, the thinning of adult males could open canopy space for sub-adults and juveniles in the understory and mid-canopy, balance sex ratios, and increase female densities while providing increased opportunities for surí production. However, this novel approach requires additional research and monitoring to test its efficacy.

While the desire to sustainably manage aguaje has not yet been fully implemented by all households, there is widespread support for adopting more sustainable practices regarding aguaje management and evidence from the 2010 harvest season suggests that these communities are willing and committed to changing management strategies. The fact that over half of the palms harvested were climbed as opposed to felled just one year after the limited training workshops, combined with the fact that over half of respondents decreased harvest intensity because they did not know how to climb and/or did not want to further degrade aguaje resources suggests increased adoption of non-destructive harvesting techniques is possible. The Maijuna have strong cultural and subsistence connections to aguaje and aguajales and therefore interest in sustainable management is high despite its current low economic importance. We suggest leveraging the cultural and socio-economic importance of aguajales to wildlife to gain momentum in sustainably managing aguaje. Hunting represents the primary source of income for the Maijuna (41%) and is an economic activity undertaken by a majority of households (93%). Focusing on the return on effort to an income stream that is currently more relevant than aguaje harvest may aid in the adoption of climbing technologies and stock improvement activities. Manzi and Coomes' (2009) findings that households with extensive hunting expertise were more likely to partake in management strategies offer support to this strategy.

Addressing market barriers will likely meet with more success as stocks improve, though this may be difficult to accomplish regardless. A more stable stock should make premium prices easier to negotiate and increase visitation of outside buyers. Increased harvest also gives economies of scale to transportation should the Maijuna attempt to avoid middlemen by transporting aguaje directly to markets in Iquitos or Mazán. Addressing market barriers will likely require stronger social organization than currently exists. However, the Maijuna may be able to build on the social capital garnered in implementing management strategies to improve female density. Even with limited success in addressing market barriers, the Maijuna should see a rise in income or return on effort from aguaje if they are able to successfully address density issues; the added income afforded by improved prices would amplify these successes.

6.4. Conclusions and implications

Given the long history of destructive harvesting in the region surrounding Iquitos and widespread reports of degraded aguajales, we believe our study communities are indicative of many communities in the Peruvian Amazon. Low female densities are likely to hinder management efforts, especially if low prices create a low return on effort. Communities that have maintained a high dependence on aguaje for income will find a greater incentive to organize and manage aguaje resources. Through the combined ecological and socio-economic efforts of our study we were able to identify the four factors limiting management success for the Maijuna. The challenges identified will also need to be addressed in a multi-disciplinary manner to realize the conservation and socio-economic benefits of sustainable *M. flexuosa* management.

Our conclusions are also relevant to the broader understanding of community-based management of NTFPs. While many NTFP studies focus on evaluating and assessing the potential of new NTFP extraction for communities, fewer studies have explored the additional challenges communities face when attempting to change the management of existing NTFP systems. Our study suggests that in cases where the resource has already been degraded, transitioning to a more sustainable management alternative can be difficult as communities face ecological and socio-economic factors that interact synergistically to hinder implementation. More research on these interactions is required as many communities and NTFPs face similar situations. Increased understanding of these factors and how they interact in varying situations would provide valuable insight to better inform the design and implementation of NTFP harvest and management.

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